Jinwoo Jeon

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Education

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
Ph.D. Candidate in Electrical Engineering — *Feb. 2023 – Present*

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
M.S. in Electrical Engineering — *Mar. 2021 – Feb. 2023*

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
B.S. in Electrical Engineering (Minor in Computer Science) — *Mar. 2016 – Feb. 2021*

Honors & Awards

2026 – 1st Place, SLAM category of Hilti × Trimble Challenge'26 in IEEE ICRA among 62 international teams
2026 – Travel Grant for IEEE ICRA'26, Institute of Control, Robotics and Systems (ICROS)

Technical Skills

Programming: C++, Python, MATLAB, CUDA
Frameworks & Tools: ROS/ROS2, PyTorch, OpenCV, Docker, Git
Research Areas: SLAM, Visual SLAM, Computer vision, Deep learning, Sensor fusion

Selected Projects

VINS-Fusion-ROS2 [★ **378**] | – Developed and maintained the first open-source ROS2 port of VINS-Fusion, supporting ROS2 execution and optional GPU-enabled feature tracking.
KAIST VIO dataset [★ **110**] *RA-L 2021* | – UAV-based visual-inertial odometry benchmark dataset for evaluating VO/VIO robustness under challenging motions, including fast motion, rotational motion, and pure rotation.

Research Project Experience

Development of a Dexterous and Agile Humanoid Robot for Industrial Environments — Urban Robotics Lab, KAIST
Supported by Korea Research Institute for defense Technology planning and advancement — *Jan. 2024 – Present*
- Led a team responsible for the development and integration of localization, terrain mapping, and path planning algorithms for autonomous humanoid navigation
- Developed a deep learning-based inertial-joint odometry algorithm for humanoid state estimation
- Keywords: Humanoid robot, Autonomous navigation, Deep inertial-joint odometry, Python, PyTorch, OpenCV, C++, ROS

Navigation & Control for Micro Swarm Robots — Urban Robotics Lab, KAIST
Supported by Korea Research Institute for defense Technology planning and advancement — *Mar. 2022 – Present*
- Was in charge of the development of visual SLAM for swarm robots
- Developed and evaluated computationally efficient visual SLAM algorithms for deployment on the NVIDIA Jetson board
- Developed tracking recovery and loop closure mechanisms to address tracking failures and accumulated trajectory drift
- Keywords: Visual SLAM, Swarm robots, Embedded system, OpenCV, C++, ROS

Depth Estimation Framework Using Mono Camera and 2D LiDAR Sensor Fusion — Urban Robotics Lab, KAIST
Supported by HYUNDAI KEFICO — *Mar. 2022 – Dec. 2022*
- Was in charge of the development of a training framework for depth completion
- Evaluated deep learning-based depth estimation model on NVIDIA Jetson platforms
- Keywords: Depth estimation, Embedded system, PyTorch, OpenCV

Autonomous drone navigation for power line inspection in underground — Urban Robotics Lab, KAIST
Supported by KEPCO — *Mar. 2021 – Jan. 2022*
- Was in charge of testing visual SLAM and supporting drone hardware assembly with sensor integration
- Conducted visual SLAM experiments to support localization and autonomous flight validation
- Keywords: Drones, Sensor setup, visual SLAM, OpenCV, C++, ROS

Quadrupedal Robots for Surveillance, Reconnaissance, and Guard Post with Improved Mobility in Challenging Terrain — Urban Robotics Lab, KAIST
Supported by Defense Agency for Technology and Quality — *Mar. 2021 – Dec. 2024*
- Was in charge of the development and evaluation of visual SLAM algorithms for quadrupedal robot navigation
- Developed visual SLAM recovery techniques to support continuous operation following state estimation failures
- Keywords: Quadruped robot, visual SLAM, OpenCV, C++, ROS

Unmanned Swarm CPS Research Lab — Urban Robotics Lab, KAIST
Supported by Defense Agency for Technology and Quality, Agency for Defense Development — *Feb. 2022 – Jan. 2023*
- Was in charge of the development of visual SLAM with relative pose estimation for unmanned swarm robotic systems
- Investigated object detection-based visual feature rejection and relative pose estimation techniques
- Ported object detection models to C++ to support efficient runtime inference
- Keywords: Visual SLAM, Object detection, Relative pose estimation, PyTorch, LibTorch, OpenCV, C++, ROS

Publications

1. **Jinwoo Jeon**, Dong-Uk Seo, Hyun Myung, "DUET-SLAM: Dual-Flow Consistency for Dense Monocular SLAM in Dynamic Environments with Geometry-aware Foundation Model," *Int. Conf. Control, Automat. and Syst. (ICCAS)*, Accepted. To appear.
2. Dong-Uk Seo, Jiwon Park, Jei Kong, **Jinwoo Jeon**, Hyun Myung, "Leveraging Planar Regularity for Gaussian Splatting SLAM," *Int. Conf. Control, Automat. and Syst. (ICCAS)*, Accepted. To appear.
3. Junwan Choi, Woongrae Jo, Dong-Uk Seo, **Jinwoo Jeon**, Hyun Myung, "Robust Structureless Monocular Visual-Inertial Initialization Exploiting Line Features and Vanishing Points," *IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Accepted. To appear.
4. Dong-Uk Seo, **Jinwoo Jeon**, Eungchang Mason Lee, Hyun Myung, "GaussianFlow SLAM: Monocular Gaussian Splatting SLAM Guided by GaussianFlow," *IEEE Robot. Automat. Lett. (RA-L)*, vol. 11, no. 6, pp. 7644-7651, 2026.
5. **Jinwoo Jeon**, Dong-Uk Seo, Eungchang Mason Lee, Hyun Myung, "AIM-SLAM: Dense Monocular SLAM via Adaptive and Informative Multi-View Keyframe Prioritization with Foundation Model," *IEEE Int. Conf. Robot. Automat. (ICRA)*, 2026.
6. K. C. Marsim, **Jinwoo Jeon**, Yeeun Kim, Myeongwoo Jeong, Hyun Myung, "CHADET: Cross-Hierarchical-Attention for Depth-Completion Using Unsupervised Lightweight Transformer," *IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, 2025, pp. 4931-4938.
7. **Jinwoo Jeon**, Myungwoo Jeong, Hyun Myung, "Fuzzy Logic and Neural Network-based Intelligent Control System for Quadruped Robot on Extreme Terrain," *Int. Conf. Control, Automat. and Syst. (ICCAS)*, 2023, pp. 1885-1889.
8. Byeongho Yu, Yeeun Kim, **Jinwoo Jeon**, I Made Aswin Nahrendra, and Hyun Myung, "A Robust State Estimator for Legged Robots using Visual-Inertial-Leg Kinematics in the Wild," *Int. Conf. Control, Automat. and Syst. (ICCAS)*, 2022.
9. Eungchang Mason Lee, **Jinwoo Jeon**, Hyun Myung, "TROT-Q: Traversability and Obstacle Aware Target Tracking System for Quadruped Robots," *ASCC*, 2022, pp. 2480-2484.
10. **Jinwoo Jeon**, Hyunjun Lim, Dong-Uk Seo, Hyun Myung, "Struct-MDC: Mesh-Refined Unsupervised Depth Completion Leveraging Structural Regularities from Visual SLAM," *IEEE Robot. Automat. Lett. (RA-L)*, vol. 7, no. 3, pp. 6391-6398, 2022.
11. Hyunjun Lim, **Jinwoo Jeon**, Hyun Myung, "UV-SLAM: Unconstrained Line-based SLAM Using Vanishing Points for Structural Mapping," *IEEE Robot. Automat. Lett. (RA-L)*, vol. 7, no. 2, pp. 1518-1525, 2022.
12. Seunghyun Lee, Sumin Chang, Eungchang Mason Lee, Junho Choi, **Jinwoo Jeon**, Seoktae Kim, Hyun Myung, "TPL: Trajectory Planner for Target Tracking in Low-light Environments," *Proc. Int. Conf. Robot Intell. Tech. Appl. (RiTA)*, 2021, pp. 12-23.
13. Eungchang Mason Lee, Dong-Uk Seo, **Jinwoo Jeon**, Hyun Myung, "QR-SCAN: Traversable Region Scan for Quadruped Robot Exploration Using Lightweight Precomputed Trajectory," *Int. Conf. Control, Automat. and Syst. (ICCAS)*, 2021, pp. 957-961.
14. ChanYoung Kim, EungChang Mason Lee, JunHo Choi, **Jinwoo Jeon**, Seoktae Kim, Hyun Myung, "ROLAND: Robust Landing of UAV on Moving Platform Using Object Detection and UWB-based Extended Kalman Filter," *Int. Conf. Control, Automat. and Syst. (ICCAS)*, 2021, pp. 249-254.
15. **Jinwoo Jeon***, Sungwook Jung*, Eungchang Lee, Duckyu Choi, Hyun Myung, "Run Your Visual-Inertial Odometry on NVIDIA Jetson: Benchmark Tests on a Micro Aerial Vehicle," *IEEE Robot. Automat. Lett. (RA-L)*, vol. 6, no. 3, pp. 5332-5339, 2021.

Experiences

Teaching Assistant

Fall 2021	EE202 Signals and Systems
Spring 2022	EE305 Introduction to Electronics Design Lab.
Spring 2023	EE381 Control System Engineering
Fall 2023	EE585 Mobile Robotics and Autonomous Navigation
Spring 2024	EE478 Introduction to Multi-disciplinary Robotics
Fall 2024	EE585 Mobile Robotics and Autonomous Navigation
Fall 2025	EE585 Mobile Robotics and Autonomous Navigation